

# Study Session 6 Checklist

Residual Diagnostics, Heteroscedasticity, and Log-Transformed Models

## Setup & Data

- ☐ Open the notebook for Session 6 (Residual Diagnostics / Heteroscedasticity).
- ☐ Load data/processed/tips\_cleaned.csv into a DataFrame df.
- ☐ Confirm numeric columns: bill\_total\_usd, tip\_usd, party\_size, tip\_pct.

## Baseline Linear Model

- ☐ Define  $X = [\text{bill\_total\_usd}, \text{party\_size}]$  and  $y = \text{tip\_usd}$ .
- ☐ Fit a baseline LinearRegression (scikit-learn) or OLS (statsmodels).
- ☐ Store predictions  $y_{\text{pred}}$  and residuals  $= y - y_{\text{pred}}$ .
- ☐ Compute baseline metrics (RMSE,  $R^2$ ) and record them in a Markdown cell.

## Residual vs Predicted Plot

- ☐ Create a scatter plot:  $x = \text{predicted tips}$ ,  $y = \text{residuals}$ .
- ☐ Add horizontal line at 0 (perfect predictions).
- ☐ Check visually: Are residuals randomly scattered (good)?
- ☐ Look for funnel shape (increasing spread) → possible heteroscedasticity.
- ☐ Save figure using save\_and\_log\_plot (if available).

## Residual Distribution & Normality

- ☐ Plot histogram + KDE of residuals (sns.histplot with kde=True).
- ☐ Create a Q–Q plot using scipy.stats.probplot or statsmodels API.
- ☐ Check visually: Do residuals roughly follow a straight line in Q–Q?
- ☐ Note deviations in the tails or skewness in a Markdown cell.

## Homoscedasticity (Equal-Variance) Checks

- ☐ Re-examine residuals vs predicted for patterns in variance.
- ☐ Optionally run Breusch–Pagan test using statsmodels.stats.diagnostic.het\_breuschpagan.
- ☐ Record BP test statistic and p-value in a Markdown cell.
- ☐ Interpret result:  $p < 0.05$  suggests heteroscedasticity.

## Log-Transformed Model (Log Tip)

- ☐ Create  $y_{\text{log}} = \text{np.log1p}(\text{tip\_usd})$  to stabilize variance.
- ☐ Fit new model:  $y_{\text{log}} \sim \text{bill\_total\_usd} + \text{party\_size}$ .
- ☐ Generate predictions on log scale and back-transform via  $\text{np.expm1}$ .
- ☐ Compute RMSE and  $R^2$  for the log model (in original tip\_usd units).
- ☐ Compare baseline vs log-model metrics in a small table.

## Residual Diagnostics for Log Model

- ☐ Plot residuals vs predicted for the log-based model.
- ☐ Plot histogram + Q–Q plot of new residuals.
- ☐ Check if variance is more constant across predicted values.
- ☐ Summarize improvements (or not) in a Markdown ELI5 explanation.

## Reflection & Storytelling

- ☐ Write a short paragraph explaining what heteroscedasticity is.

## Study Session 6 Checklist (cont.)

- ☐ Explain why log-transforming `tip_usd` can help stabilize variance.
- ☐ Describe which model you would report to a stakeholder and why.